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(1) Research Program/Project/Study Title: "Enhancing Beverage Quality: Utilization of Matured Coconut Water and Pineapple Concentrate with Seaweed Extract in Functional Drinks"	
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E-mail Address:	

Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	Roxas City
		Province:	Capiz
		Region:	6
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	

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(7) Sector/Commodity	(8) Discipline/Field of Specialization
(9) Rationale Food waste is a critical global issue, with implications spanning economic, environmental, and social domains. Throughout the food supply chain, from production to consumption, significant amounts of food are lost or discarded, exacerbating inefficiencies and sustainability challenges. To address this issue, innovative approaches are imperative, leveraging underutilized resources and promoting sustainable practices. In the Philippines, the coconut industry stands as a cornerstone of the economy, producing valuable commodities such as virgin coconut oil, coconut milk, and dried coconut. However, a substantial byproduct of this process is mature coconut water, often overlooked and discarded as waste due to prevalent processing methods. This overabundance of mature coconut water poses environmental concerns, necessitating strategies for its utilization to mitigate waste and maximize resource efficiency. Other agricultural products, like pineapple, offer opportunities for value addition and waste reduction. Pineapple, renowned for its nutrient density and health benefits, presents versatile applications, from dietary consumption to industrial uses such as food coloring. Moreover, seafood, particularly seaweed, holds immense potential for enhancing nutritional value and functional properties in food and beverage products. Seaweed is a rich source of essential vitamins, minerals, and bioactive compounds, offering numerous health benefits ranging from improved digestion to enhanced immune function. Furthermore, seaweed cultivation has minimal environmental impact, making it a sustainable option for ingredient sourcing. In light of these considerations, this study explores the utilization of matured coconut water and pineapple concentrate, enriched with seaweed extract, in the development of functional drinks. By integrating these underutilized resources and harnessing the nutritional benefits of seaweed, this research aims to reduce food waste, promote sustainable resource utilization, and enhance the nutritional quality of functional beverages. This research aims to offer practical solutions for addressing food waste, promoting sustainable agriculture, and advancing the nutritional value of functional beverages with a particular focus on the utilization of seaweed, mature coconut water and pineapple with its health-promoting properties and environmental sustainability.	
(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)	

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(10) Objectives/Statement of the Problem

The research focus is to determine the acceptability of Matured Coconut water in making Pineapple Juice Concentrate with seaweed extract.

1.To determine the most acceptable treatment of matured coconut water and pineapple concentrate with seaweed extract.

2.To determine the sensorial characteristics of the most acceptable treatment of matured coconut water and pineapple concentrate with seaweed extract.

3.To determine the physico- chemical characteristics (pH and TSS) and microbial load (yeast and mold count) of the most acceptable treatment of matured coconut water and pineapple concentrate with seaweed extract after incubation period.

4.To determine the consumer acceptable of the most acceptable treatment of matured coconut water and pineapple concentrate with seaweed extract.

(11) Review of Literature/Prior Art Search

2.1 Matured coconut water

Coconut (*Cocos nucifera* L.) is an important fruit tree found in tropical regions and its fruit can be made into a variety of foods and beverages. Normally people in tropical countries drink tender coconut water like a beverage as a sports drink. However, mature coconut water (MCW) is normally discharged as only the coconut meat is used for producing coconut milk. For instance, in Thailand MCW is a by-product from the coconut milk industry that might be used to produce nata decoco (Cappelletti et al., 2015).

According to (F. Ismail N. A. Et al.,2019). Matured coconut water is an agro waste that produced mainly from coconut milk industry which contains variety of sugars and minerals. This study was carried out on synthesizing levulinic acid from matured coconut water by focusing on the effect of catalytic acid to coconut water ratio, types of acid and reaction time.

Coconut water may be an old commodity which is usually consumed as fresh, pasteurized coconut water (PCW) becomes a new entrant in the Philippine beverage industry. As a new player, penetrating the market of the giant beverage manufacturers seem to be risky and challenging. The study

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aimed to determine the market potential of bottled pasteurized coconut water and to identify product positioning opportunity in the beverage market. The study employed random consumption survey and product test. (Hanilyn A. Hidalgo 2017).

2.2 MCW: Health Benefits

May benefit to manage diabetes- Research shows that Coconut water may help to manage blood sugar levels and improve diabetes symptoms. It is a good source of manganese, which may increase insulin sensitivity and reduce blood sugar levels. **Help in good digestion-** Coconut water is rich in manganese that helps to regulate your bowel movements. It may prevent gaseous distension of the abdomen, constipation and acidity. **Reduce fatigue and stress-** There are instances such as fatigue, stress where you feel lethargic and don't want to eat or do anything. (Dr.Nikita Toshi 2023).

2.3 Pineapple

Pineapple (*Ananas comosus*) is a large tropical fruit with a spiky, tough skin and sweet insides. When European explorers came across it in South America in the 17th century, they called it a pineapple because of its pinecone appearance. The scaly bumps on the outside are called "eyes." Slice it open, and you'll find bright yellow flesh that's both sweet and tart. For centuries, pineapples were so rare that only very wealthy people could afford to buy them. Some people even rented the exotic fruit. (Mikstas Christine, RD, LD 2022).

Pineapple, scientifically known as *Ananas Comosus* Merr., is one of the country's most popular fruits. It is also one of the top earners of foreign exchange among the country's agricultural commodities. The fruit contains water, carbohydrates, substantial amount of Vitamin C and potassium, and other nutrients. (agrimag 2019).

2.4 Pineapple: Health Benefits

May Reduce Blood Cholesterol. A preliminary review of several tropical fruits describes lower serum lipid profiles, triglycerides, and low-density lipoprotein (LDL) in rats who were given pineapple extract every day for 30 days. The authors also noted better cardiovascular health and lower inflammation as a result.

Decreased Inflammation. Like other types of fruit, pineapple is high in antioxidants, a feature that may reduce inflammation in the body (Cherney kristeen et al., 2022).

2.5 Concentrate. Is made from mostly fresh fruit. Fruit concentrate is produced from filtering the water out of fruit pulp. The concentrate product is stored under aseptic, non-aseptic and frozen conditions. These methods have been historically used to optimize its transportation across the world, to then be re-constituted with the same volume of water that was evaporated from it, once it reaches the location of juice manufacture. (Ogunsola Adeoti 2020).

2.6 Hot Filling. Hot-fill is a packaging and processing technique in which heat is used to sterilize the food product and subsequently the inner surface of the container during the filling process. It is a technique that is used extensively in the food packaging industry for a variety of applications ranging from applesauce to dips and sauces. The hot-fill process starts by initially heating the food product up to a temperature between 194-204°F as a method of killing bacteria and microorganisms present in the food (Moren Michael 2020).

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

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(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and

Procedure

The following were the steps followed in the preparation of the mature coconut water:

Preparation of the matured coconut water:

1. Collect coconut water and filter it.
2. Boil for 10 minutes and set aside.

Preparation of pineapple:

1. Peel the pineapple and remove the pineapple surface.
2. Cut the pineapple into cubes.
3. Put the pineapple in a blender and blend until pureed.
4. Set aside.

Preparation of seaweed extract:

Below are the steps followed in the preparation of seaweed extract:

1. Wash seaweed thoroughly with running water five times to remove adherent dirt and debris.
2. Drain for 5 minutes and weigh.
3. The seaweeds will be soaked in 6 cups of boiling water together with 1 cup of calamansi extract for about 1 hour and 30 minutes.
4. Strain to separate the seaweed extract.

Preparation of matured coconut water and pineapple juice concentrate with seaweed extract.

1. In a casserole, put the coconut water and sugar, then boil for 10 minutes.
2. Add the pineapple puree and seaweed extract and bring to a boil for about 15 minutes.
3. Remove the concentrate from the casserole and hot-fill it into the glass bottle.

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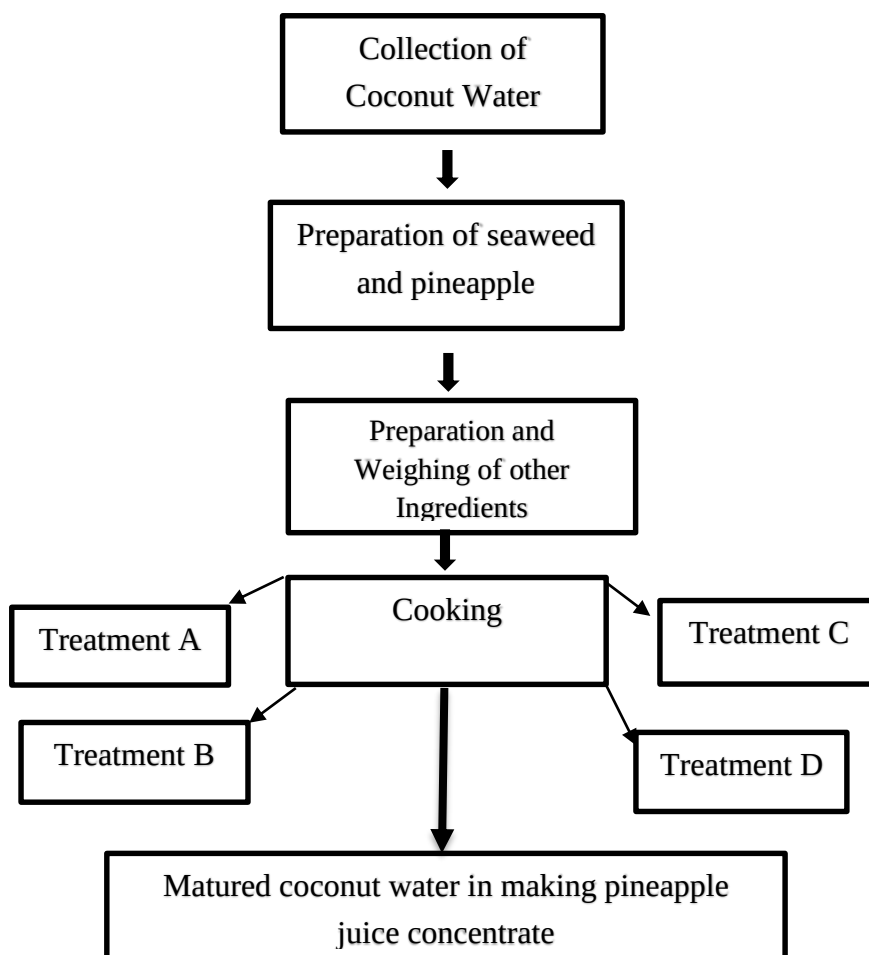


Figure 1. Flow Chart in Acceptability of Matured Coconut Water in Making Pineapple Juice Concentrate

Experimental Method and Design

This study was a 3-way factor with three (3) treatments. In this study, the researcher develops and formulated her own recipe of every product for evaluation. The amount of coconut water, pineapple extract and seaweed extract will be manipulated in every treatments.

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(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

A ready to go drink that helps reduce health disorders with its different properties, a healthy alternative beverage instead of unhealthy beverages sold in the market

(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Process will be taught to the target clientele through extension program of the college, however, prior to the extension activity, formulation of the beverage will be copyrighted.

(17) Target Beneficiaries (Who the clienteles are and what are the expected outcome/effects of the use of the projects output)

SME's , entrepreneurs and farmers who are into beverage industry

(18) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(19) Budgetary Requirement

Sources of Expenses	Quantity	Amount (Php)
• Coconut water	50 lit.	1000
• Glacelaria Seaweed	1 sack	1000
• Pineapple fruit	10 kg	2000
• Sugar	20kg.	2000
• Lemon/Calamansi	5 kg.	1000
• Sensory Plastic Cups with cover	200 cups	1000
• Packaging materials	14 pcs./box glass bottle@25	2000
• Physico chemical test	Ph/tss	1000
• Microbial analysis	Yeast and mold count	3000

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• Nutrifacts		10,000
• Transportation Allowance		2000
• Ph meter		70,000
Total		P95,000.00

(20) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

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(21) Capsule Curriculum Vitae (One-page brief C.V. only)

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Academic history

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Doctoral **Capiz State University Main Campus, Fuentes Drive, Roxas City, 2020**
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Master's **Isabela State University, Education Department**
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